Indian Statistical Institute

Final Examination 2025-2026

Analysis of Several Variables, B.Math Second Year

Time : 3 Hours Date : 07.11.2025 Maximum Marks : 100 Instructor : Jaydeb Sarkar

---

Q1. (20 marks) Let C be a simple closed smooth curve that lies in the plane

$$x + y + z = 1.$$

Compute

$$\int_C z\,dx - 2x\,dy + 3y\,dz,$$

in terms of the area of the region enclosed by C.

Q2. (20 marks) Determine whether or not $F = \langle z \sec^2 x, z, y + \tan x \rangle$ is conservative. If F is conservative, find a potential function for F.

Q3. (20 marks) Define $f : \mathbb{R}^2 \to \mathbb{R}$ by

$$f(x,y) = \begin{cases} (x^2+y^2)\sin\left(\frac{1}{\sqrt{x^2+y^2}}\right) & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$$

Which of the following statements are true (justify your answer)?

(1) f is differentiable at $(0,0)$.

(2) f is continuous at $(0,0)$.

(3) f_x and f_y are not continuous at $(0,0)$.

Q4. (20 marks) Prove Green's theorem for the special case

$$\mathcal{R} = \{(x,y) : a \leq x \leq b, f(x) \leq y \leq g(x)\},$$

where f and g are continuous functions.

Q5. (20 marks) Use an appropriate change of variables to evaluate

$$\int_R \sin(x+y)\cos(x-y)\,dA,$$

where R is the triangle with vertices $(0,0)$, (π,π), and $(\pi,-\pi)$.

Q6. (10 marks) Compute the flux of the vector field $F = \langle x, y, z \rangle$ over the part of the cylinder $x^2 + y^2 = 4$ that lies between the planes $z = 0$ and $z = 2$ with normal pointing away from the origin.